NEUROSCI/PSY 217: Introduction to Cognitive Neuroscience

Syllabus

Instructor
Kevin S. LaBar, Ph.D. (he, him)
"Professor LaBar" or "Dr. LaBar" are OK
Professor, Center for Cognitive Neuroscience
Room B247, LSRC Building
Tel: (919) 681-0664
E-mail: klabar@duke.edu

Teaching Assistants
Jaime Rios and Jim Zhang
Graduate Students, Center for Cognitive Neuroscience
Room B241, LSRC Building
E-mail: jaime.rios@duke.edu, jim.zhang@duke.edu

Class lecture: location and time
Soc Sci 136 MW 3:05 - 4:20 pm

Recitation sections: locations and times

Section 01D: LSRC D243*	Th 8:45 – 9:35 am
Section 02D: LSRC A247*	Th 3:20 – 4:10 pm
Section 03D: LSRC A156*	F 10:20 – 11:10 am
Section 04D: LSRC A156*	F 3:20 – 4:10 pm

*Note that the Neuroscience Teaching Labs indicated on the syllabus will be held at the Duke Institute for Brain Sciences (DIBS) Glass Cube in LSRC (go down to lowest level in the cube and turn left)

Office hours
Right after class/recitation or by e-mail appointment to Prof. LaBar and TAs Rios and Zhang.

Course pre-requisites
Students should feel sufficiently comfortable with neuroanatomy, neuroscience methods, and cognitive psychology to read primary journal articles or review chapters in the field of cognitive neuroscience. Students should have taken NEUROSCI 101/PSY 106, NEUROSCI 102/PSY 107, PSY 102, or NEUROSCI 195FS/NEUROBIO 195FS prior to enrolling in this course. This course is not intended for first-year students who have had no prior college-level neuroscience or psychology course.

Learning goals
To understand how cognitive functions are instantiated in neural systems of the brain and to appreciate the value of different methods in the field. Students should learn the implications of cognitive neuroscience for understanding dysfunction in clinical disorders, as well as some ethical and professionalism issues in conducting cognitive neuroscience research. Students will learn how neuroscience functions are implemented at the circuit/systems level of analysis, primarily in humans with some additional coverage of non-human primates. Students will become familiar with various specialty topics that span anatomy, neuroscience methods, sensation

and perception, motor control, attention, memory, emotion, social cognition, language, executive function, and decision making. In the discussion sections, students will learn to critically evaluate current research in the field, familiarize themselves with anatomy, behavioral tasks and methods that are commonly used, and contribute to debates about conceptual topics as well as current research practices.

Course format

- Lectures led by Dr. LaBar will be accompanied by PowerPoint slides that cover the main concepts to be learned in the course. These slides will be provided on the course Canvas site to facilitate understanding of the reading material. Some demos and exercises conducted in class will not be included in the slides. Students will also co-present an article in class to provide an in-depth example of a topic covered by the lecture material, and they will be expected to post a slide summary to Canvas as well.
- Recitation sections led by the TA will include discussion of primary readings and review articles, break-out group activities including active learning experiences with various cognitive tasks and research methods, and commentary about popular media coverage, scientific practices, or ethical principles.

Required materials for class and recitation sections

No textbook is required, but a copy of Marie Banich and Rebecca Compton's *Cognitive Neuroscience* textbook has been placed on reserve in the library as a resource for students. It is recommended that students spend an hour or two each week to familiarize themselves with this resource, as it contextualizes and expands upon what was covered in class. Additional readings and other ancillary material will be posted on the Canvas site for the course and will be free to download as .pdf files. Software for neuroscience labs will be made available on the computers in the DIBS Cube lab space.

Grading

Grades will be based on the following criteria:

25 % -- first examination
25 % -- second examination
25 % -- third examination
15 % -- class participation (includes participation in lectures and recitations)
10 % -- in-class paper or recitation presentation

Because the TA is responsible for the initial grading of material for the course, if a student has a question about the grading, they should first contact the TA. If the matter is not resolved to their satisfaction, then the student should contact Dr. LaBar and he will meet separately with the student and TA to resolve the grading issue.

Course requirements

Class Attendance and Preparation: Attendance is important because there is no required textbook, and some exercises or information provided in class and in recitation will not be covered by the lecture Powerpoint slides. It is expected that each student has read the material assigned for the lecture or recitation section prior to the start of class and that students will contribute to class discussions and participate in the active learning activities during recitation sections.

Exams: Three in-class, non-cumulative exams will cover core concepts covered in lectures and recitation sections. Topics covered in the PowerPoint slides/discussion in class or recitation will be emphasized over other details of the assigned readings. Learning objectives will be posted on Canvas and review sessions will be given in recitation sections prior to each exam. Be prepared to bring questions to the review sessions. Each exam will constitute approximately 30 questions in a mixed format (multiple-choice, short answer, definitions, matching, true/false).

Presentation of current papers: Students will lead class discussion in pairs on one primary literature paper that covers current issues on the topics that align with the lecture portion of the class. These oral presentations should be accompanied by slides that will be posted on the course's Canvas site and should both engage the class in an interesting discussion as well as summarize key points in the readings. The slides should be uploaded to Canvas prior to the start of class. Because there are not sufficient numbers of slots to include everyone in the lecture portion of class, some students will be assigned to lead a discussion in their recitation section instead.

Make-up policy This course abides by the Trinity College policies on incapacitating illnesses (https://trinity.duke.edu/undergraduate/academic-policies/illness) for exams, in-class or discussion section presentations that you were assigned to lead, and recitation attendance. Please note that short-term illness forms are not needed for occasional missed lecture attendance, as there is nothing to 'make-up' for not attending them. Due to their active learning and interactive discussion components, attendance will be taken for recitation sections, and recitation attendance will be incorporated into the class participation grade. Note that this policy indicates that STINFs are only permitted for *incapacitating* illnesses or injuries. An incapacitation notification form must be completed and submitted to Dr. LaBar 1 hour prior to the missed class or recitation section. If you have a long-term illness or chronic condition that may affect your academic performance, notify your academic dean, and he or she will provide notification to course professors detailing any accommodations that should be made. Please contact your TA or another student in the course to obtain information about what was presented during class/recitation sections and not included in the PowerPoint slides posted on-line.

- Exam make-ups: All make-up exams will consist of 5 essay questions. Make-up exams are only permitted if an Incapacitation Form or Academic Deans' excuse for long-term illness is submitted
  1 hour before the start of class on the scheduled exam date. Make-up exams take place at 8:00 am two days after the scheduled exam date in Professor LaBar's office (Room B247, LSRC Building). The make-up exam for Exam I will be administered at 8:00 am on Wednesday 9/24; the make-up exam for Exam II will be administered at 8:00 am on Wednesday 10/29; the make-up exam for Exam III will be administered at 8:00 am on Friday 12/5. Any student who fails to show up for an exam without an Incapacitation Form or Academic Dean's notification of a long-term illness will receive a grade of 'F' for the exam. Only 1 exam STINF will be accepted. Any subsequent STINFs submitted for exams will not be accepted and a grade of 'F' will be given for the exam.

- Paper presentation make-ups: If a STINF is submitted on the day a student was assigned to present a paper in class or recitation, then the student should make arrangements to have their portion of the slide presentation covered by the co-presenter, if possible. Within 3 days of the missed assignment, the student will have to upload to Canvas a recorded video of themselves covering the material they were supposed to present to the class (using zoom, for instance). If the video is not posted within 3 days and/or no slides have been prepared for the co-presenter in class, then the student will receive a grade of 'F' for their presentation. If no STINF is provided, the student will receive a grade of 'F' for their presentation.

Special accommodations policy

Students with special needs can receive accommodations for their in-class work only if their Academic Dean submits an official accommodations notification to Professor LaBar. Varsity student-athletes must also provide official notification for excused absences at the beginning of the semester to Professor LaBar. Varsity athletes can arrange to make up missed material in advance of the due date; in this case, the exam will not be all-essay.

Cellphone, laptop, audiovisual recording, and note-sharing policies

No texting, web browsing, or personal cellphone use is permitted in class. Psychological research has shown that these activities are detrimental to student learning and are a distraction to others, including the professors and TAs, in class. See: https://www.brookings.edu/research/for-better-learning-in-college-lectures-lay-down-the-laptop-and-pick-up-a-pen/. Students are permitted to use laptops only to take notes or download resources from the Canvas website and are encouraged to use traditional pen-and-paper to take notes as this activity requires more deliberate note-taking in the students' own words, which facilitates memory encoding. No audio or visual recording is permitted in class with the exception of special needs accommodations. No portion of the class notes or lectures can be freely publicly shared or uploaded to on-line sources for a fee, as the professor retains the copyright to the intellectual material in the lectures, and other resources provided (textbook materials, figures/illustrations, additional readings) are also copyright-protected.

Academic integrity policy

Students are reminded to conduct themselves with academic integrity in accordance with the Duke Community Standard (http://www.integrity.duke.edu).

Use of AI Policy

Student Policy: The use of AI assistance for exams is strictly prohibited. The use of AI assistance for presentations is permitted ONLY for generating ideas, conducting general background research, or basic editing of already-written material. No verbal or pictorial material generated from a generative AI source (e.g., ChatGPT) is permitted to be included in the presentations themselves. Any AI use in preparation for oral presentation assignments must be fully documented. Along with your slides for the presentations, please submit documentation of all interactions with AI assistants, including screen shots that capture the interactions with the assistant, a written rationale for why you think it was appropriate for you to use the assistant in the way that you did, and documentation of how you changed verbiage/pictures generated by the AI agent before including it in your presentation. Please be aware that AI assistants are notorious for generating false information, including creating fake scientific references. If it is suspected that AI was used to assist with exams or that its outputs were included in presentations, you will receive an "F" grade on the assignment and will be referred to disciplinary action for academic dishonesty. Asking an AI assistant to complete the work for you is equivalent to asking another human being to complete the work for you.

Instructor/TA Policy: Neither the instructor nor the TA for this course will use generative AI for grading purposes or for providing feedback to students on their assignments.

Academic and Health Resources

Students are reminded that Duke offers free consultations and peer tutoring services through the Academic Resource Center. Students who are concerned about their well-being during the course of the semester can take advantage of the resources available through Counseling and Psychological Services (CAPS), Student Health and other campus organizations (see Duke Reach https://studentaffairs.duke.edu/dukereach, Duke Line https://sites.duke.edu/dukeline or CAPS https://students.duke.edu/wellness/caps).

Readings and assignments

DATE	TOPIC	READING ASSIGNMENT All assigned readings are posted on Canvas, supplemental textbook readings available on reserve in the Duke Library	PAPER PRESENTERS
	UNIT I. METHODS, PERCEPTION, ACTION		
M 8/25: Lecture	Course Overview	--	--
W 8/27: Lecture	Functional Anatomy	Purves et al., 2012 Appendix	
Th-F 8/28-8/29: Recitation	Scientific Reading and Presentation	Carey 2020; Naegle, 2021	--
M 9/1: Lecture	NO CLASS	LABOR DAY	--
W 9/3: Lecture	Methods I	Purves et al., 2012 Ch. 2 (pp. 17-29); Poldrack & Farah, 2015	Jaime Rios
Th-F 9/4-5: Recitation	Brain Anatomy Demo	Strotzer, 2009	--
M 9/8: Lecture	Methods II	Purves et al., 2012 Ch. 2 (pp. 29-53); da Silva, 2013	Jim Zhang
W 9/10: Lecture	Vision	Op de Beeck et al., 2019	Sofia Gonzalez and Dan Jellinek
Th-F 9/11-9/12: Recitation	Neurosci Teaching Lab: Alpha Waves Demo (*in the DIBS Cube)		--
M 9/15: Lecture	Audition	Sihvonen et al. 2022	Sophia Becraft and Nam Ho
W 9/17: Lecture	Motor Systems	Ebrahimi et al., 2024	Amy Colocho and Eujin Chung
Th-F 9/18-19: Recitation	Perception / Exam Review	The Dress article and paper	Thurs AM: Nathan Shenkerman; Thurs PM: Anthony Ayala and Rachel Koo; Fri AM: Fabiana Henriquez and Kareena Sukhanan; Fri PM: N/A
M 9/22: Lecture	EXAM I	--	--
	UNIT II. ATTENTION, MEMORY & EMOTION		
W 9/24: Lecture	Attention	Cai et al., 2024	Kristina Schaufele and Melanie Perez-Romero
Th-F 9/25-9/26 Recitation	Neurosci Teaching Lab: P300 Demo (*in the DIBS Cube)		--
M 9/29: Lecture	Attentional Control	Van Vleet et al., 2020	Clara Sun and Emi Takayama
W 10/1: Lecture	Nondeclarative Memory	Palombo et al., 2021	Reese Woodworth and Sungmin Jung

Th-F 10/2-10/3: Recitation	Attention		Thurs AM: N/A; Thurs PM: Justine Medveckus; Fri AM: N/A; Fri PM: N/A
M 10/6: Lecture	Declarative Memory	Barrett et al., 2024	Oluwadurotimi Akinyelu and Vyom Patel
W 10/8: Lecture	Emotion Theory	Cowen et al., 2019	Elizabeth Buduen and Kaining Zhao
Th-F 10/9-10/10: Recitation	Memory		--
M 10/13: Lecture	NO CLASS	FALL BREAK	--
W 10/15: Lecture	Fear Conditioning	Borgomaneri et al., 2020	Skye Flores and Chisa Ihekwereme
Th-F 10/16-10/17 Recitation	Emotion I		--
M 10/20: Lecture	Emotional Memory	Young et al., 2017	J'Vonae Fitchett and Praise Oladiti
W 10/22: Lecture	Emotion Regulation	Bellaiche et al., 2025	Maria Romero Degwitz and Allie Wolff
Th-F 10/23-10/24 Recitation	Emotion II/Exam Review		--
M 10/27: Lecture	EXAM II		--
UNIT III: SOCIAL AND EXECUTIVE FUNCTIONS ACROSS THE LIFESPAN			
W 10/29: Lecture	Language	Card et al., 2024; Chang, 2024	Alex Benz and Erin Lim
Th-F 10/30-10/31 Recitation	Language		--
M 11/3: Lecture	Social Cognition I	Wang et al., 2020	Ella Cariello and Olaya Garcia-Grau
W 11/5: Lecture	Social Cognition II	Pyasik et al., 2023	Jocelyn Perez-Ugarte and Nicolas Tadros
Th-F 11/6-11/7: Recitation	Social Cognition		--
M 11/10: Lecture	Executive Function I	Jones & Graff-Radford, 2021	Natalie Cuevas and Renee Li
W 11/12: Lecture	Executive Function II	Lloyd et al., 2021	Nadine Fanous and Sanjit Pamidi
Th-F 11/13-11/14 Recitation	Executive Function		--
M 11/17: Lecture	NO CLASS	SOCIETY FOR NEUROSCIENCE MEETING	--
W 11/19: Lecture	Decision Making	Spaniol et al., 2018	Katherine Li and Ashmi Trivedi
Th-F 11/20-11/21 Recitation	Decision Making/Exam III review		--
M 11/24: Lecture	Development	Szenczy et al., 2025	Bradley Barrett and Maria Junaid

W 11/26: Lecture	NO CLASS	THANKSGIVING BREAK	--
Th-F 11/27-11/28 Recitation	NO RECITATION	THANKSGIVING BREAK	--
M 12/1: Lecture	Aging	Fields et al., 2022	Alexa Reid and Gabi Graneiro
W 12/3: Lecture	EXAM III	--	--
Th-F 12/4-12/5: Recitation	NO RECITATION	END OF SEMESTER	--